Member 3

Shwetha S

PES1UG24CS452

Section 4: Requirements Traceability Matrix (RTM)

4.1 Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) establishes a relationship between system requirements, test cases, and validation methods. It ensures that every requirement can be verified and that no requirement is missed during testing.

Requirement ID	Requirement	Type	Test Case	Validation Method
FR-01	The system shall allow the user to start a quiz.	FR	TC-01	Functional Testing
FR-02	The system shall display multiple-choice questions.	FR	TC-02	Functional Testing
FR-03	The system shall allow the user to select an answer.	FR	TC-03	Functional Testing
FR-04	The system shall allow the user to submit the quiz.	FR	TC-04	Functional Testing
FR-05	The system shall calculate the user's score.	FR	TC-05	Functional Testing
FR-06	The system shall display the quiz result to the user.	FR	TC-06	Functional Testing
FR-07	The system shall allow the administrator to log in.	FR	TC-07	Functional Testing
FR-08	The system shall allow the administrator to add questions.	FR	TC-08	Functional Testing
FR-09	The system shall allow the administrator to edit questions.	FR	TC-09	Functional Testing
FR-10	The system shall allow the administrator to delete questions.	FR	TC-10	Functional Testing
VFR-01	The system shall reject invalid or empty answer inputs.	VFR	TC-11	Input Validation / Boundary Testing

Requirement ID	Requirement	Type	Test Case	Validation Method
VFR-02	The system shall prevent unauthorized users from accessing administrator functions.	VFR	TC-12	Security Testing
VFR-03	The system shall prevent submission of incomplete required information.	VFR	TC-13	Input Validation
VFR-04	The system shall reject invalid administrator login credentials.	VFR	TC-14	Security Testing
VFR-05	The system shall display appropriate error messages for invalid inputs.	VFR	TC-15	Usability / Input Validation

4.2 Test Case Summary

Test Case ID	Test Case Description	Expected Result
TC-01	Click on "Start Quiz".	The quiz should start successfully.
TC-02	Start a quiz and view the questions.	Multiple-choice questions should be displayed.
TC-03	Select an answer for a question.	The selected answer should be recorded.
TC-04	Click on "Submit Quiz".	The quiz should be submitted successfully.
TC-05	Submit a completed quiz.	The system should calculate the correct score.
TC-06	Complete the quiz.	The score/result should be displayed.
TC-07	Enter valid administrator credentials.	The administrator should be logged in.
TC-08	Add a new question.	The question should be added successfully.
TC-09	Edit an existing question.	The question should be updated successfully.
TC-10	Delete an existing question.	The question should be deleted successfully.
TC-11	Submit without providing a required answer.	The system should reject the input and display a validation message.
TC-12	Attempt to access administrator functions without authorization.	Access should be denied.
TC-13	Submit incomplete required information.	The system should reject the submission.

Test Case ID	Test Case Description	Expected Result
TC-14	Enter incorrect administrator login credentials.	Login should fail and an appropriate message should be displayed.
TC-15	Enter invalid input in a form field.	The system should reject or appropriately handle the input.

4.3 Detailed Test Cases

Detailed test cases are prepared to verify whether the Online Quiz System satisfies its functional and validation requirements. Each test case specifies the preconditions, test steps, test data and expected results. The actual results and status will be recorded during the testing phase.

4.3.1 Functional Test Cases

Test Case ID	**TC-01**
Test Case Name	**Start Quiz**
Requirement ID	**FR-01**
Preconditions	**Application is running and a quiz is available.**
Test Steps	**1. Open the application. 2. Click on "Start Quiz".**
Test Data	**Available quiz**
Expected Result	**The quiz should start and display the first question.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-02**
Test Case Name	**Display Questions**
Requirement ID	**FR-02**
Preconditions	**A quiz has been started.**
Test Steps	**1. Start a quiz. 2. Observe the displayed question and answer options.**
Test Data	**Available quiz questions**

Expected Result	**Multiple-choice questions and their answer options should be displayed correctly.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-03**
Test Case Name	**Select Answer**
Requirement ID	**FR-03**
Preconditions	**A question and its options are displayed.**
Test Steps	**1. Select one answer option. 2. Check whether the selection is recorded.**
Test Data	**One valid answer option**
Expected Result	**The selected answer should be recorded successfully.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-04**
Test Case Name	**Navigate Questions**
Requirement ID	**FR-04**
Preconditions	**A quiz has been started.**
Test Steps	**1. Click the Next button. 2. Click the Previous button.**
Test Data	**Available quiz questions**
Expected Result	**The system should allow the user to move between questions correctly.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-05**
Test Case Name	**Submit Quiz**
Requirement ID	**FR-05**
Preconditions	**A quiz has been started.**
Test Steps	**1. Answer the questions. 2. Click "Submit Quiz".**
Test Data	**Completed quiz**
Expected Result	**The quiz should be submitted successfully.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-06**
Test Case Name	**Calculate Score**
Requirement ID	**FR-06**
Preconditions	**A quiz has been submitted.**
Test Steps	**1. Submit answers with known correct and incorrect responses. 2. Check the calculated score.**
Test Data	**Known correct and incorrect answers**
Expected Result	**The system should calculate the score correctly.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-07**
Test Case Name	**Display Result**
Requirement ID	**FR-07**
Preconditions	**The quiz has been submitted and evaluated.**
Test Steps	**1. Complete the quiz. 2. Submit it. 3. Observe the result page.**
Test Data	**Completed quiz**
Expected Result	**The user's score and result should be displayed.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-08**
Test Case Name	**Exit Quiz**
Requirement ID	**FR-08**
Preconditions	**A quiz is in progress.**
Test Steps	**1. Click the "Exit Quiz" option. 2. Observe the system response.**
Test Data	**Active quiz**
Expected Result	**The user should be able to exit the quiz successfully.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

4.3.2 Administrator Functional Test Cases

Test Case ID	**TC-09**
Test Case Name	**Administrator Login**
Requirement ID	**FR-09**
Preconditions	**Administrator account exists.**
Test Steps	**1. Open the admin login page. 2. Enter valid credentials. 3. Click Login.**
Test Data	**Valid administrator credentials**
Expected Result	**Administrator should be logged in successfully.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-10**
Test Case Name	**Add Question**
Requirement ID	**FR-10**
Preconditions	**Administrator is logged in.**
Test Steps	**1. Open question management. 2. Enter question details and options. 3. Save the question.**
Test Data	**Valid question with answer options**
Expected Result	**The new question should be added successfully.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-11**
Test Case Name	**Edit Question**
Requirement ID	**FR-11**
Preconditions	**Administrator is logged in and a question exists.**
Test Steps	**1. Select an existing question. 2. Modify its details. 3. Save changes.**
Test Data	**Updated question details**
Expected Result	**The question should be updated successfully.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-12**
Test Case Name	**Delete Question**
Requirement ID	**FR-12**
Preconditions	**Administrator is logged in and a question exists.**
Test Steps	**1. Select an existing question. 2. Click Delete. 3. Confirm deletion if prompted.**
Test Data	**Existing question**
Expected Result	**The selected question should be deleted successfully.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-13**
Test Case Name	**View Questions**
Requirement ID	**FR-13**
Preconditions	**Administrator is logged in and questions are stored.**
Test Steps	**1. Open the question management page. 2. View the available questions.**
Test Data	**Stored quiz questions**
Expected Result	**The system should display the stored questions correctly.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

4.3.3 Input Validation and Security Test Cases

Test Case ID	**TC-14**
Test Case Name	**Invalid Answer Validation**
Requirement ID	**VFR-01**
Preconditions	**A quiz question is displayed.**
Test Steps	**Attempt to submit an answer that is not among the provided options.**
Test Data	**Invalid answer value**
Expected Result	**The system should reject the invalid answer.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-15**
Test Case Name	**Unanswered Question Validation**
Requirement ID	**VFR-02**
Preconditions	**A quiz is in progress with at least one unanswered question.**
Test Steps	**Attempt to submit the quiz without answering all required questions.**
Test Data	**Quiz containing unanswered questions**
Expected Result	**The system should identify unanswered questions and display a warning according to quiz rules.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-16**
Test Case Name	**Unauthorized Admin Access**
Requirement ID	**VFR-03**
Preconditions	**User is not logged in as an administrator.**
Test Steps	**Attempt to access an administrator page directly.**
Test Data	**Unauthenticated user**
Expected Result	**Access to administrator functions should be denied.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-17**
Test Case Name	**Invalid Admin Login**
Requirement ID	**VFR-04**
Preconditions	**Administrator login page is available.**
Test Steps	**1. Enter incorrect credentials. 2. Click Login.**
Test Data	**Invalid username or password**
Expected Result	**Login should fail and an appropriate error message should be displayed.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-18**
Test Case Name	**Question Input Validation**
Requirement ID	**VFR-05**
Preconditions	**Administrator is logged in.**
Test Steps	**Attempt to add a question with missing or invalid required information.**
Test Data	**Incomplete question and missing options**
Expected Result	**The system should reject incomplete or invalid question information.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

4.3.4 Verification and Non-Functional Test Cases

Test Case ID	**TC-19**
Test Case Name	**Score Calculation Verification**
Requirement ID	**VFR-06**
Preconditions	**Quiz contains questions with known correct answers.**
Test Steps	**Submit a set of answers with a predetermined number of correct responses and compare the result.**
Test Data	**For example, 4 correct answers out of 5 questions.**
Expected Result	**The calculated score should match the expected score.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-20**
Test Case Name	**Data Integrity Verification**
Requirement ID	**VFR-07**
Preconditions	**Administrator is logged in and questions exist.**
Test Steps	**Add, edit and delete questions. Check whether the stored question data remains consistent.**
Test Data	**Existing and modified questions**
Expected Result	**Question management operations should not cause inconsistent quiz data.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-21**
Test Case Name	**Stored Data Validation**
Requirement ID	**VFR-08**
Preconditions	**Administrator is logged in.**
Test Steps	**Attempt to save question information containing invalid or incomplete data.**
Test Data	**Invalid question and answer data**
Expected Result	**Invalid data should be rejected before database storage.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-22**
Test Case Name	**Session Security Validation**
Requirement ID	**VFR-09**
Preconditions	**Administrator has logged in and subsequently logged out.**
Test Steps	**Attempt to access a protected admin page after logout.**
Test Data	**Logged-out administrator session**
Expected Result	**Access to protected pages should be denied.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-23**
Test Case Name	**Error Handling Verification**
Requirement ID	**VFR-10**
Preconditions	**Application is running.**
Test Steps	**Enter invalid data or perform an invalid operation.**
Test Data	**Invalid form input**
Expected Result	**The system should display a suitable error message.**

Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-24**
Test Case Name	**Usability Validation**
Requirement ID	**VFR-11**
Preconditions	**Application is accessible.**
Test Steps	**Navigate through the quiz and administrator interfaces and observe their clarity and ease of use.**
Test Data	**Normal user interactions**
Expected Result	**The interface should be understandable and easy to navigate.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Test Case ID	**TC-25**
Test Case Name	**Performance Validation**
Requirement ID	**VFR-12**
Preconditions	**Application is running under normal expected load.**
Test Steps	**Open quiz pages and load questions while observing response time.**
Test Data	**Normal quiz usage**
Expected Result	**The application should respond within a reasonable time under normal load.**
Actual Result	**To be recorded during testing.**
Status	**Not Run**

Section 5: Validation Plan

The validation plan defines the testing activities that will be carried out to ensure that the Online Quiz System satisfies its specified requirements and behaves correctly under valid and invalid conditions. Since Phase 1 focuses on planning, the testing activities described below are planned for subsequent phases.

5.1 Functional Testing

Functional testing will verify that all major functions of the Online Quiz System operate according to the specified requirements.

The following functions will be tested:

- Starting a quiz
- Displaying multiple-choice questions
- Selecting answers
- Submitting a quiz
- Calculating the score
- Displaying quiz results
- Administrator login
- Adding questions
- Editing questions
- Deleting questions

5.2 Non-Functional Testing

Non-functional testing will evaluate the quality attributes of the system.

The following aspects will be considered:

- **Performance:** Verify that quiz pages and questions load within an acceptable time.
- **Usability:** Verify that users can easily navigate through the quiz and understand the interface.
- **Reliability:** Verify that the system performs consistently without unexpected failures.
- **Compatibility:** Verify that the system functions correctly on supported browsers and devices.

5.3 Security Testing

Security testing will verify that the system protects restricted functionality and prevents unauthorized access.

The following areas will be tested:

- Administrator authentication
- Unauthorized access to administrator functions

- Invalid administrator login attempts
- User and administrator access control
- Session management
- Protection of quiz and administrator data

For example, if a normal quiz participant attempts to access an administrator page without authorization, the system should deny access.

5.4 Input Validation Testing

Input validation testing will verify that the system correctly handles invalid, empty, or incomplete inputs.

The following cases will be tested:

- Empty answer submission
- Empty question fields
- Missing answer options
- Invalid login credentials
- Invalid or unexpected input values
- Incomplete form submissions

The system should reject invalid inputs and display appropriate validation messages

5.5 Usability Testing

Usability testing will verify that the Online Quiz System is simple and easy to use.

The following aspects will be evaluated:

- Clear navigation
- Clearly labelled buttons
- Readable questions and answer options
- Simple quiz submission process
- Easy access to quiz results
- Clear and understandable error messages

A first-time user should be able to start, complete, and submit a quiz without requiring technical assistance.

5.6 Basic Penetration-Testing Plan

Basic penetration testing will be planned to identify common security vulnerabilities in the Online Quiz System. The testing will focus on authentication, authorization, input handling, and session management.

The following security checks are planned:

1. **Unauthorized Access:** Attempt to access administrator pages without logging in. Access should be denied.
2. **Invalid Login Attempts:** Attempt to log in using incorrect credentials. The system should reject the login attempt.
3. **Input Manipulation:** Enter unexpected or malicious input into form fields. The system should safely handle or reject such input.
4. **Session Security:** Attempt to access protected pages after logging out. Access should be denied.
5. **Access Control:** Verify that normal users cannot perform administrator operations.